

Project Development Phase Performance Testing

Date	4 September 2026
Team ID	ServiceNow
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	5 Marks

Objective:

To verify that the Automated Network Request Management application performs reliably under expected and peak load — covering response time, concurrent user handling, and stability over extended use — for its actual scope of Service Catalog, Approval Workflow, and Notifications (no CMDB, external integration, or chatbot component).

Test Environment & Tools:

- Environment: ServiceNow UAT/sub-production instance (<https://<instance-name>.service-now.com>)
- Load/Performance tools: Apache JMeter (portal load simulation), ServiceNow Performance Analytics (dashboard/report timing)

Performance Test Scenarios & Results

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
PT-01	Catalog Form Load Time	Measure the time taken for the Network Request catalog item to fully load on the Service Portal.	Form should load in under 2 seconds.	Average load time measured at 1.6 seconds across 10 runs.	Pass
PT-02	Request Submission Response Time	Use a timer to measure the time from clicking "Submit" to receiving the confirmation screen.	Should complete in under 3 seconds.	Average submission time measured at 2.1 seconds across 10 runs.	Pass
PT-03	Concurrent User Load (Service Portal)	Simulate 100 concurrent users submitting network requests at the same time.	No significant degradation; response time increase should stay under 20%.	Response time increased by 15% under 100 concurrent users; no errors observed.	Pass
PT-04	Approval Workflow Processing Time	Measure the time from request submission to the approval task being created and assigned.	Approval task should be created within 5 seconds.	Approval task created in an average of 3.4 seconds.	Pass
PT-05	Notification Delivery Time	Measure the time from an approval/rejection action to the requester receiving an email/Teams notification.	Notification should be delivered within 1 minute.	Email delivered in ~20 seconds; Teams notification delivered in ~45 seconds.	Pass
PT-06	SLA Dashboard Load Time	Measure dashboard render time with 5,000 open/closed request records in the system.	Dashboard should render in under 3 seconds.	Dashboard rendered in an average of 2.7 seconds.	Pass
PT-07	File Attachment Upload Performance	Upload supporting documents of varying sizes (1MB, 5MB, 10MB) to a request record.	Uploads should complete without timeout up to 10MB per file.	All file sizes uploaded successfully without timeout or data loss.	Pass
PT-08	Stress Test — Peak Load	Simulate 3x normal peak concurrent request submissions (300 users) for 15 minutes.	System should remain stable, queuing requests gracefully rather than crashing or timing out.	System remained stable; requests queued with a temporary response delay of ~4 seconds, no crashes.	Pass

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
PT-09	Scalability Test — Growing Data Volume	Run report and dashboard queries against a dataset of 10,000+ historical request records.	Queries and dashboards should perform within acceptable time as data volume grows.	Report generation time increased from 2s to 3.5s at 10,000 records; still within acceptable range.	Pass
PT-10	Soak / Endurance Test	Run the application under normal expected load continuously for 24 hours.	No memory leaks, performance degradation, or unexpected errors over the test period.	No memory leaks observed; response times remained consistent throughout the 24-hour run.	Pass

Performance Benchmark Summary

Performance Metric	Target	Actual (Observed)	Status
Catalog form load time	< 2 sec	1.6 sec	Met
Request submission response time	< 3 sec	2.1 sec	Met
Approval task creation time	< 5 sec	3.4 sec	Met
Notification delivery time	< 1 min	~45 sec (worst case)	Met
SLA dashboard render time (5,000 records)	< 3 sec	2.7 sec	Met
Concurrent user degradation (100 users)	< 20% increase	15% increase	Met

Observations & Recommendations:

- All core portal, approval workflow, notification, and dashboard metrics met their targets under both normal and peak load.
- The system remained stable under 3x peak concurrent load (300 users) and showed no memory leaks or degradation over a 24-hour soak test.
- Since fulfilment is handled manually by the network team (no auto-provisioning or external system integration in this project's scope), actual end-to-end request completion time depends on team responsiveness rather than application performance, and is tracked separately via the SLA dashboard rather than as a load-test metric.
- No performance issues were identified; the application is within target on every measured metric.